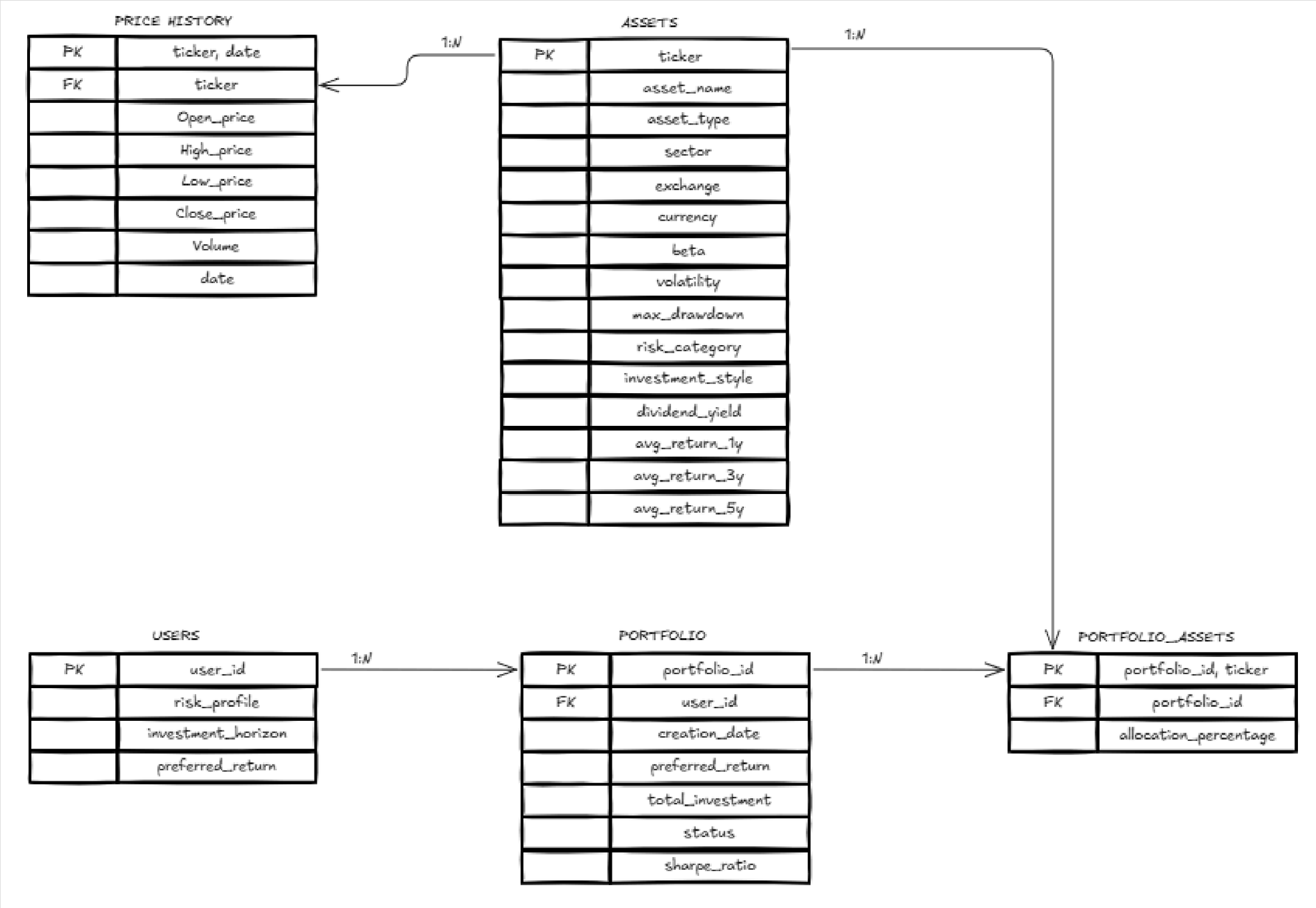
DATABASE SCHEMA



USERS

USERS

PK	user_id
	risk_profile
	investment_horizon
	preferred_return

EXAMPLE:

- user_id: USR-9921
- risk_profile: Moderate
- investment_horizon: 10 (Plans to withdraw the funds in 10 years)
- preferred_return: 8.0 (Targets an average annual return of 8%)

ASSETS

PK	ticker
	asset_name
	asset_type
	sector
	exchange
	currency
	beta
	volatility
	max_drawdown
	risk_category
	investment_style
	dividend_yield
	avg_return_1y
	avg_return_3y
	avg_return_5y

ASSETS

EXAMPLE:

- ticker / asset_ticker / asset_name: TSLA / Tesla, Inc.
- asset_type: Stock
- sector: Consumer Cyclical / Automotive
- exchange: NASDAQ
- currency: USD
- beta: 2.42 (This means it is significantly more volatile than the market)
- volatility: 45% (Annualized fluctuation)
- max_drawdown: -70% (Its worst historical drop from a peak)
- risk_category: High
- investment_style: Growth (Aggressive growth)
- dividend_yield: 0.0% (Tesla typically does not pay dividends)
- avg_return_1y: 12.5%

ASSETS

PK	ticker
	asset_name
	asset_type
	sector
	exchange
	currency
	beta
	volatility
	max_drawdown
	risk_category
	investment_style
	dividend_yield
	avg_return_1y
	avg_return_3y
	avg_return_5y

WHAT IS BETA?

MATCHES THE RISK PROFILE
CONSERVATIVE CLIENT → LOWER BETA ASSETS
AGGRESSIVE CLIENT → HIGHER BETA ASSETS

- Beta = 1 → Moves in line with the market
- Beta > 1 → More volatile than the market (higher risk, higher potential return)
- Beta < 1 → Less volatile than the market (more defensive)

PRICE HISTORY

PK	ticker, date
FK	ticker
	Open_price
	High_price
	Low_price
	Close_price
	Volume
	date

PRICE_HISTORY

EXAMPLE

- ticker: AAPL (Apple), MSFT (Microsoft)
- date: 2024-02-15
- open_price: 182.35
- high_price: 184.10
- low_price: 181.90
- close_price: 183.60
- volume: 52,000,000 (shares traded that day)

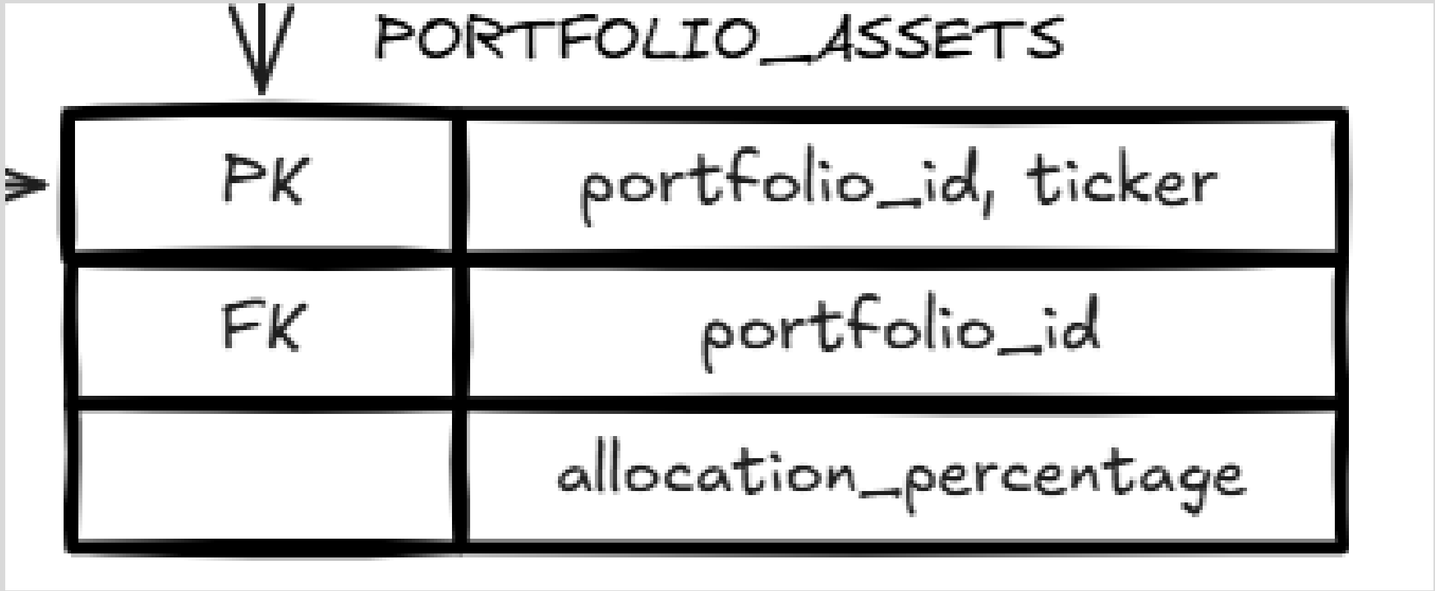
PORTFOLIO

PK	portfolio_id
FK	user_id
	creation_date
	preferred_return
	total_investment
	status
	sharpe_ratio

PORTFOLIO

EXAMPLE:

- portfolio_id: PORT-001
- user_id: USR-9921
- creation_date: 2024-02-19
- preferred_return: 8.5 (The return estimated when the portfolio was created)
- total_investment: 10,000 (Euros or Dollars)
- status: Active
- sharpe_ratio: 1.2 (Indicates a good risk–return trade-off)



PORTFOLIO_ASSETS

- portfolio_id: PORT-001.
- ticker: VTI (A US Total Stock Market ETF).
- allocation_percentage: 60.0 (60% of the user's money is allocated to this asset).